In [1]:
```
import numpy as np
import matplotlib.pyplot as plt
def plot(*funcs, x_min=-5, x_max=5):
    plt.grid(color='gray', linestyle='-', linewidth=1)
    xs = np.linspace(x_min, x_max, 50)
    for f in funcs:
        plt.plot(xs, np.vectorize(f)(xs))
```

Tag 4

39

a)

$$
\begin{aligned}
8*(x-1)-5*(y+1)&=12 && (1)\\
13*(x+2)-6*(y+2)&=61 && (2)\\
& && (3)\\
\Leftrightarrow 8x-8-5y-5&=12|+13 && (4)\\
\Leftrightarrow 13x+26-6y-12&=61|-14 && (5)\\
& && (6)\\
\Leftrightarrow 8x-5y&=25|*6 && (7)\\
\Leftrightarrow 13x-6y&=47|*5 && (8)\\
& && (9)\\
\Leftrightarrow 48x-30y&=150 && (10)\\
\Leftrightarrow 65x-30y&=235|- && (11)\\
& && (12)\\
\Leftrightarrow -17x&=-85|:(-17) && (13)\\
\Leftrightarrow x&=5 && (14)\\
& && (15)\\
8x-5y&=25|x=5 && (16)\\
& && (17)\\
\Leftrightarrow 8*(5)-5y&=25|-8*5 && (18)\\
& && (19)\\
\Leftrightarrow -5y&=-15|:(-5) && (20)\\
& && (21)\\
\Leftrightarrow y&=3 && (22)
\end{aligned}
$$

3.1.1 Eine lineare Gleichung

1

Eine Großmutter ist 84 Jahre alt, ihre Enkelin ist 8 Jahre alt.

a)

In wie vielen Jahren wird die Großmutter fünf mal so alt wie die Enkelin sein?

$84 + j = 5 * (8 + j)$

$\Leftrightarrow 84 + j = 40 + 5j| - j - 40$

$\Leftrightarrow 44 = 4j|/4$

$\Leftrightarrow 11 = j$

b)

Vor wie vielen Jahren war sie zwanzigmal so alt?

$84 - j = 20 * (8 - j)$

$\Leftrightarrow 84 - j = 160 - 20j| + 20j - 84$

$\Leftrightarrow 19j = 76|/19$

$\Leftrightarrow j = 4$

2

a)

Gegeben Rechteck mit $a * b = A$ mit $a > b$

$$
\begin{aligned}
(a - 2\text{ cm}) * (b + 2\text{ cm}) &= a * b + 4\text{ cm}^2 && (23)\\
(a + 3\text{ cm}) * (b + 3\text{ cm}) &= a * b + 57\text{ cm}^2 && (24)\\
& && (25)\\
\Leftrightarrow ab + 2a\text{ cm} - 2b\text{ cm} - 4\text{ cm}^2 &= a * b + 4\text{ cm}^2| - ab && (26)\\
\Leftrightarrow ab + 3a\text{ cm} + 3b\text{ cm} + 9\text{ cm}^2 &= a * b + 57\text{ cm}^2| - ab && (27)\\
& && (28)\\
\Leftrightarrow 2a\text{ cm} - 2b\text{ cm} - 4\text{ cm}^2 &= 4\text{ cm}^2|/2 && (29)\\
\Leftrightarrow 3a\text{ cm} + 3b\text{ cm} + 9\text{ cm}^2 &= 57\text{ cm}^2|/3 && (30)\\
& && (31)\\
\Leftrightarrow a\text{ cm} - b\text{ cm} - 2\text{ cm}^2 &= 2\text{ cm}^2| + 2\text{ cm}^2 && (32)\\
\Leftrightarrow a\text{ cm} + b\text{ cm} + 3\text{ cm}^2 &= 19\text{ cm}^2| - 3\text{ cm}^2 && (33)\\
& && (34)\\
\Leftrightarrow a\text{ cm} - b\text{ cm} &= 4\text{ cm}^2 && (35)\\
\Leftrightarrow a\text{ cm} + b\text{ cm} &= 16\text{ cm}^2 && (36)
\end{aligned}
$$

Addieren bzw. subtrahieren

$2a\text{ cm} = 20\text{ cm}^2|/2\text{ cm}$

$\Leftrightarrow a = 10$

$-2b\text{ cm} = (4 - 16)\text{ cm}^2|/2\text{ cm}$

$\Leftrightarrow b = 6$

Die Ergebnisse erfüllen $a > b$. Damit ist dies tatsächlich die Lösung.

b)

$$
\begin{aligned}
2a + b &= 37 \text{ cm} && (37)\\
b &= a - 5 \text{ cm} && (38)\\
& && (39)\\
2a + (a - 5 \text{ cm}) &= 37 \text{ cm} && (40)\\
\Leftrightarrow 2a + a - 5 \text{ cm} &= 37 \text{ cm}| + 5 \text{ cm} && (41)\\
\Leftrightarrow 3a &= 42 \text{ cm}|3 && (42)\\
\Leftrightarrow a &= 14 \text{ cm} && (43)\\
& && (44)\\
b &= a - 5 \text{ cm} && (45)\\
\Leftrightarrow b &= 14 \text{ cm} - 5 \text{ cm} && (46)\\
\Leftrightarrow b &= 9 \text{ cm} && (47)
\end{aligned}
$$

Erfüllt die Lösung die Dreiecksungleichung?

$a + a > b$
$\Leftrightarrow 14 \text{ cm} + 14 \text{ cm} > 9 \text{ cm}$

$a + b > a$
$\Leftrightarrow 14 \text{ cm} + 9 \text{ cm} > 14 \text{ cm}$

Ja

3

$$
\begin{aligned}
r &= 3l && (48)\\
3(r - 30) &= l + 30 && (49)\\
& && (50)\\
3(3l - 30) &= l + 30 && (51)\\
\Leftrightarrow 3 * 3l - 3 * 30 &= l + 30 && (52)\\
\Leftrightarrow 9l - 90 &= l + 30| - l + 90 && (53)\\
\Leftrightarrow 8l &= 120|/8 && (54)\\
\Leftrightarrow l &= 15 && (55)\\
& && (56)\\
r &= 3l && (57)\\
\Leftrightarrow r &= 3 * 15 && (58)\\
\Leftrightarrow r &= 45 && (59)
\end{aligned}
$$

Rechts sind auch mindestens dreißig Nüsse. Daher ist die Lösung gültig.

4

$$m + 16 = 2(t + 16) \tag{60}$$
$$m + t = 40| - t \tag{61}$$
$$\tag{62}$$
$$\Leftrightarrow m + 16 = 2(t + 16) \tag{63}$$
$$\Leftrightarrow m = 40 - t \tag{64}$$
$$\tag{65}$$
$$(40 - t) + 16 = 2(t + 16) \tag{66}$$
$$\Leftrightarrow 40 - t + 16 = 2t + 2 * 16 \tag{67}$$
$$\Leftrightarrow 56 - t = 2t + 32| - 56 - 2t \tag{68}$$
$$\Leftrightarrow -3t = -24|/(-3) \tag{69}$$
$$\Leftrightarrow t = 8 \tag{70}$$
$$\tag{71}$$
$$m = 40 - t \tag{72}$$
$$\Leftrightarrow m = 40 - 8 \tag{73}$$
$$\Leftrightarrow m = 32 \tag{74}$$

5

a)

$$a + b = 25 \tag{75}$$
$$a - b = 7 \tag{76}$$

Additionsverfahren

$$2a = 25 + 7 \tag{77}$$
$$\Leftrightarrow 2a = 32|/2 \tag{78}$$
$$\Leftrightarrow a = 16 \tag{79}$$

$$a - b = 7 \tag{80}$$
$$\Leftrightarrow 16 - b = 7| + b - 7 \tag{81}$$
$$\Leftrightarrow 9 = b \tag{82}$$

b)

$$4a - 3b = 18| * 3 \tag{83}$$
$$3a + 10 = 14b| * 4 \tag{84}$$
$$\tag{85}$$
$$\Leftrightarrow 12a - 9b = 54 \tag{86}$$
$$\Leftrightarrow 12a + 40 = 56b \tag{87}$$

Subtrahieren (Additionsverfahren)

$$-9b - 40 = 54 - 56b| + 56b + 40 \tag{88}$$
$$\Leftrightarrow 47b = 94|/47 \tag{89}$$
$$\Leftrightarrow b = 2 \tag{90}$$

$$
\begin{aligned}
3a + 10 &= 14b| - 10 && (91)\\
\Leftrightarrow 3a &= 14 * 2 - 10 && (92)\\
\Leftrightarrow 3a &= 18|/3 && (93)\\
\Leftrightarrow a &= 6 && (94)
\end{aligned}
$$

In [2]:
```
120/8
```

Out[2]: 15.0

In [3]:
```
%%latex
$\max_{t\in T}w(t)$
```

$\max_{t\in T} w(t)$

In [4]:
```
%%latex
\begin{align}
a + b &= 3a \\
a - b &= 2b | +b \\
\\
a &= 3b \\
3b + b &= 3(3b)| - 9b\\
-5b &= 0 | /(-5) \\
\\
b &= 0\\
\\
a + 0 &= 3a | -3a\\
-2a &= 0 | /(-2) \\
a &= 0
\end{align}
```

$$
\begin{aligned}
a + b &= 3a && (95)\\
a - b &= 2b| + b && (96)\\
& && (97)\\
a &= 3b && (98)\\
3b + b &= 3(3b)| - 9b && (99)\\
-5b &= 0|/(-5) && (100)\\
& && (101)\\
b &= 0 && (102)\\
& && (103)\\
a + 0 &= 3a| - 3a && (104)\\
-2a &= 0|/(-2) && (105)\\
a &= 0 && (106)
\end{aligned}
$$

In [5]:
```
%%latex
\begin{align}
2a + b &= 17 | *5 \\
10a + 5b &= 85 \\
\\
10a + 5b &= 85 \\
\end{align}
```

$$2a + b = 17| * 5 \tag{107}$$
$$10a + 5b = 85 \tag{108}$$
$$\tag{109}$$
$$10a + 5b = 85 \tag{110}$$

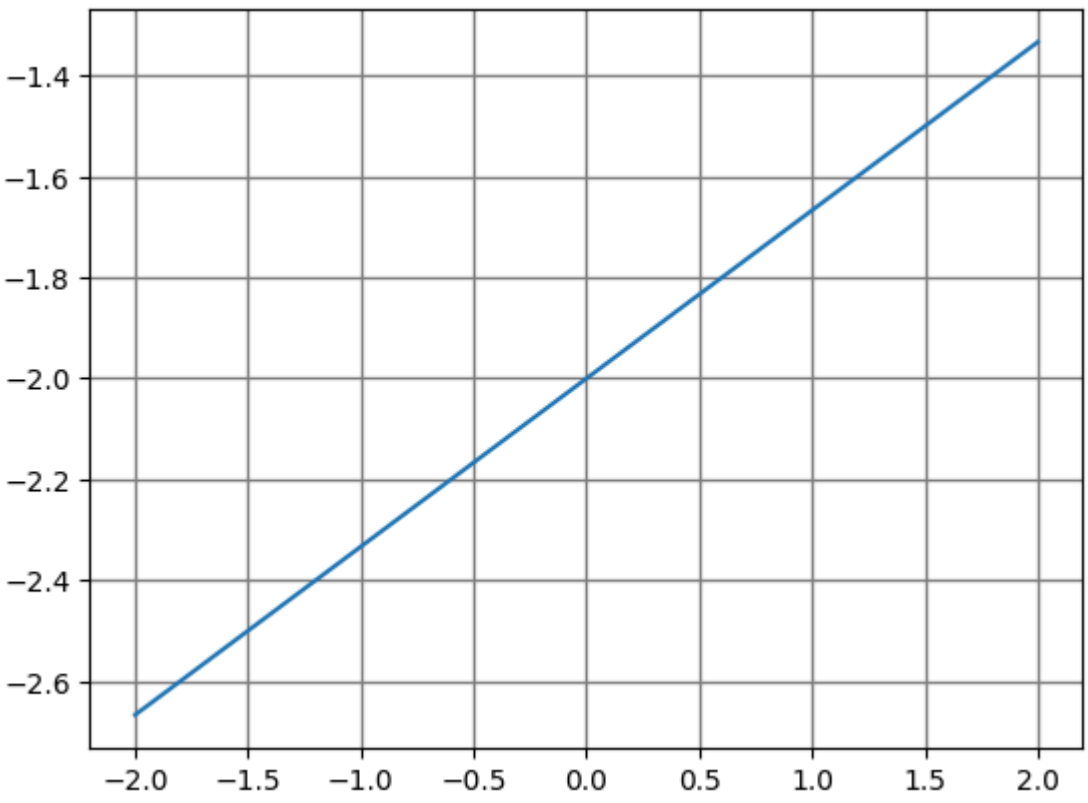
In [6]:
```
%%latex
$a= x$
```

$a = x$

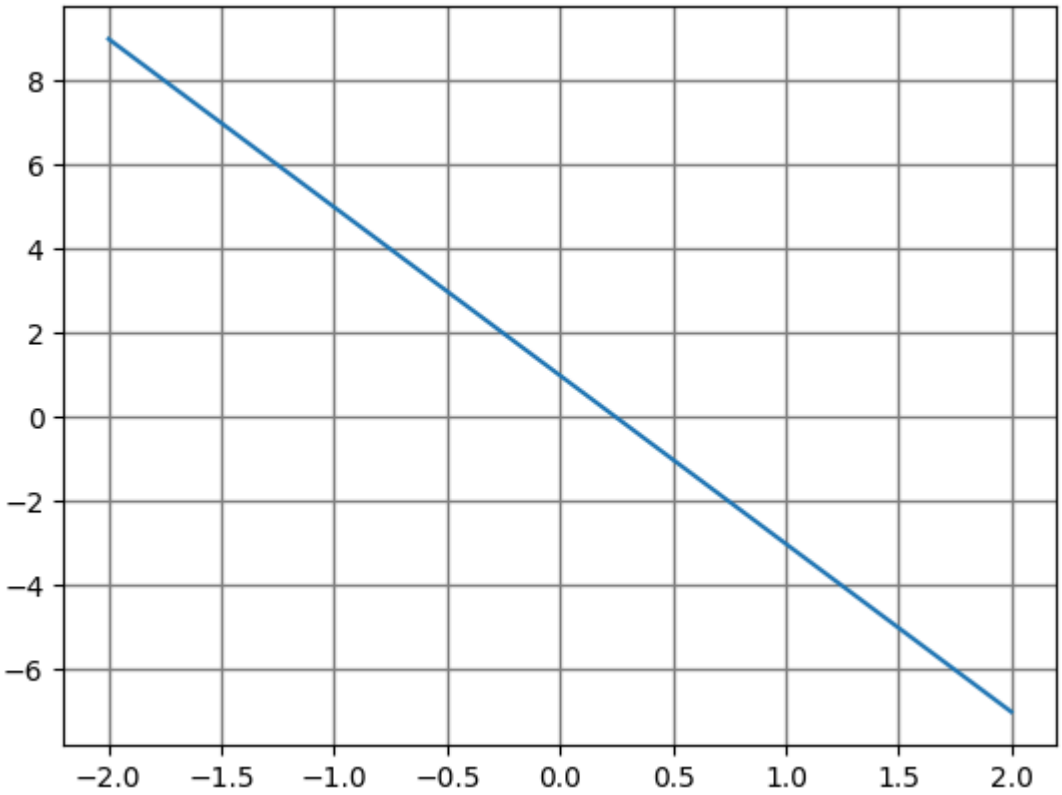
In [7]:
```
%%latex
\begin{align}
2x + b &= 17 | -2x \\
b &= 17 - 2x
\end{align}
```

$$2x + b = 17| - 2x \tag{111}$$
$$b = 17 - 2x \tag{112}$$

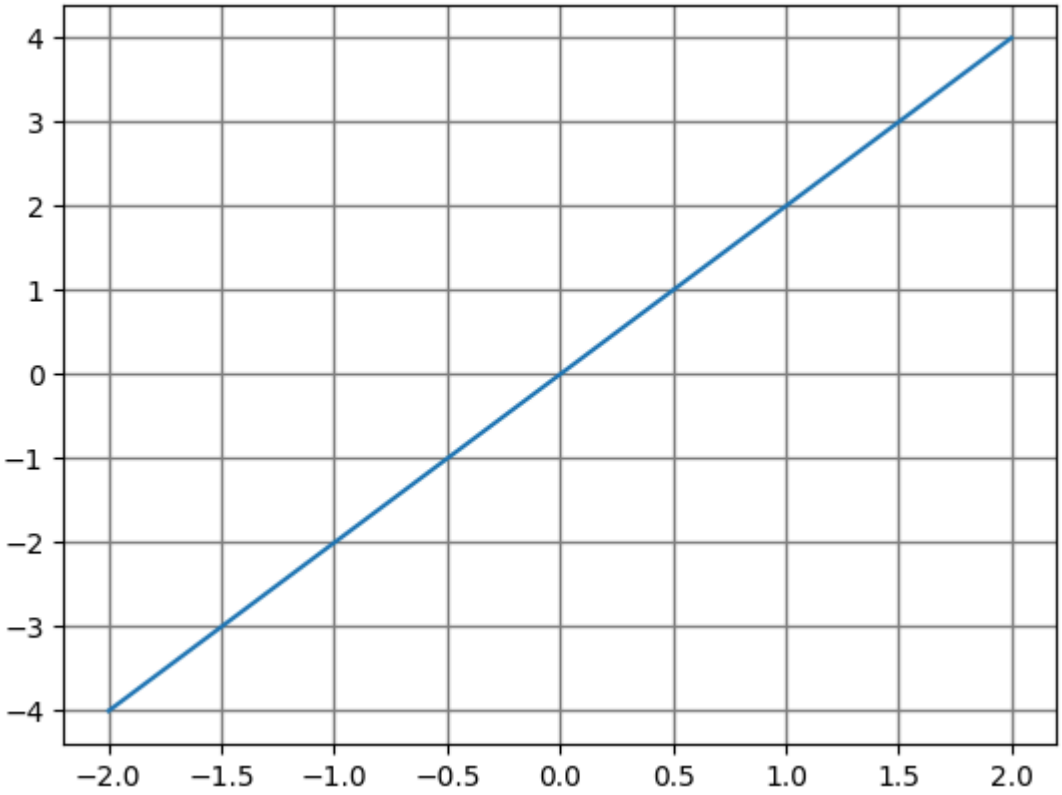
In [8]:
```
plot(lambda x: x/3-2, x_min = -2, x_max = 2)
```



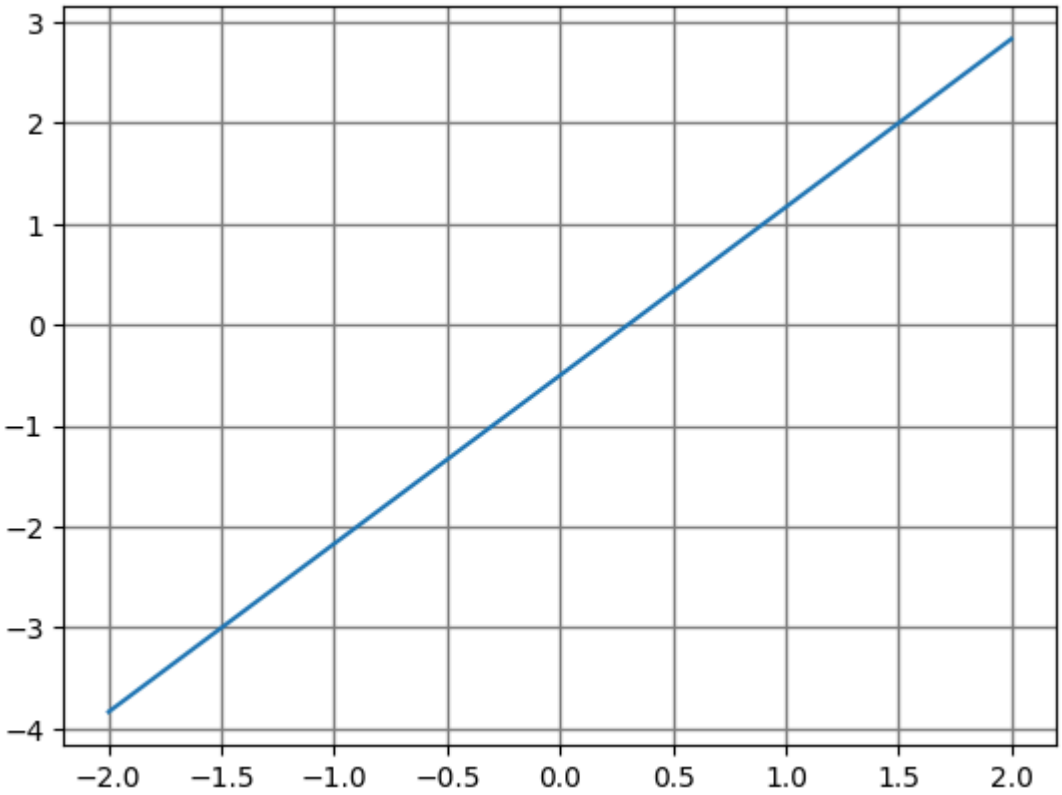
In [9]:
```
plot(lambda x: -4*x+1, x_min = -2, x_max = 2)
```



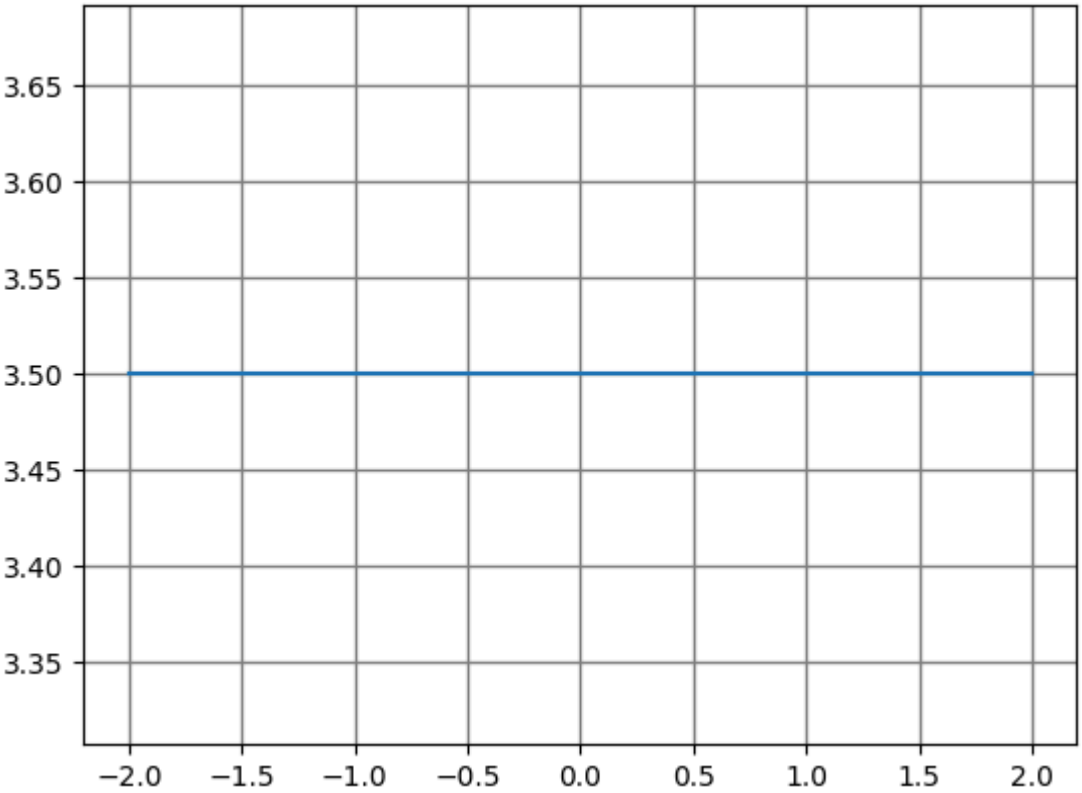
In [10]:
```
plot(lambda x: 2*x, x_min = -2, x_max = 2)
```



In [11]:
```
plot(lambda x: 5/3*x-0.5, x_min = -2, x_max = 2)
```

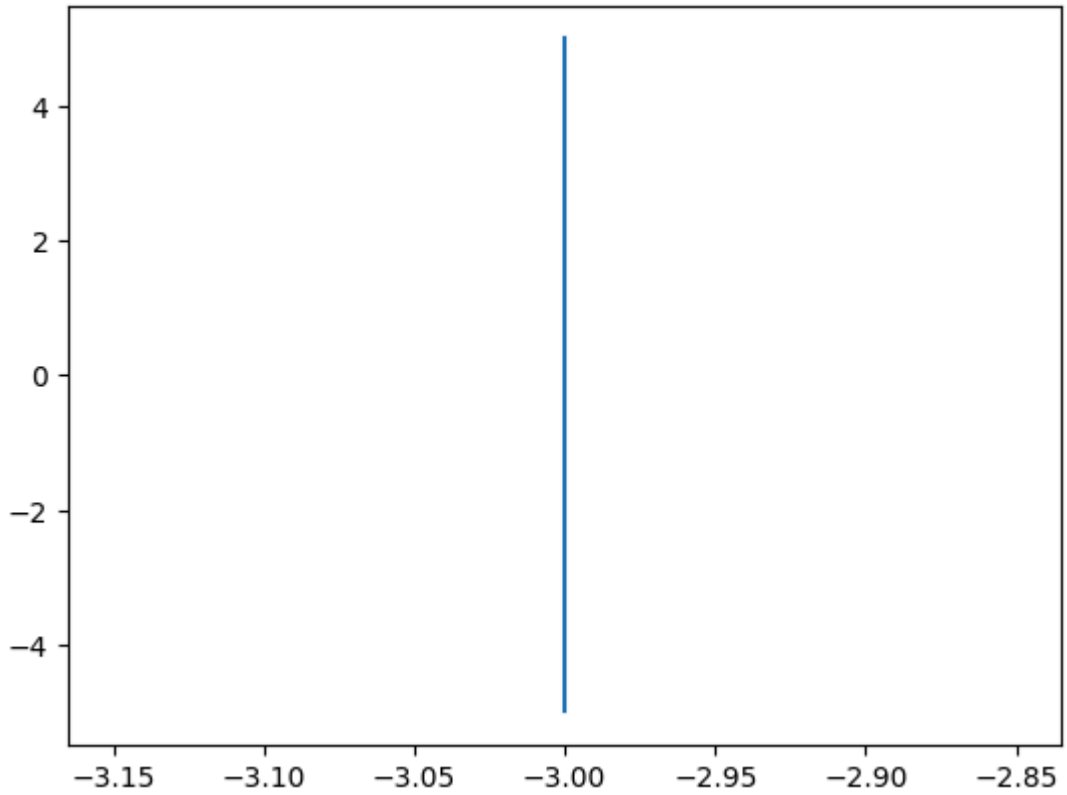


In [12]:
```
plot(lambda x: 3.5, x_min = -2, x_max = 2)
```



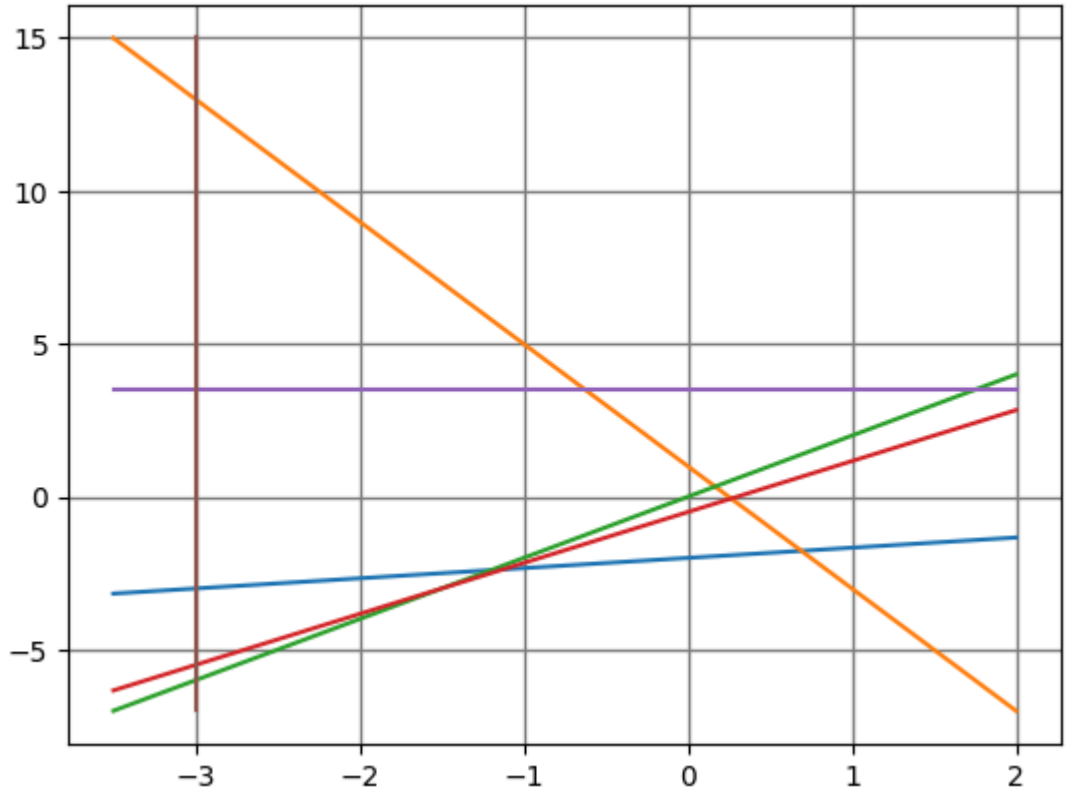
In [13]:
```
plt.plot([-3,-3],[-5,5])
```

Out[13]:
```
[<matplotlib.lines.Line2D at 0x1d87fe90110>]
```



In [14]:
```
plot(lambda x: x/3-2, lambda x: -4*x+1, lambda x: 2*x, lambda x: 5/3*x-0.5, lamb
plt.plot([-3,-3],[-7,15])
```

Out[14]:
```
[<matplotlib.lines.Line2D at 0x1d8232d35f0>]
```



In [15]:
```
%%latex
\begin{align}
A &= \frac{1}{2}h_cc \\
A + 7 &= \frac{1}{2}(h_c+3)c \\
A - 3 &= \frac{1}{2}(h_c-1)c \\
```

```
\\
\frac{1}{2}h_cc + 7 &= \frac{1}{2}(h_c+3)c |*2 \\
\frac{1}{2}h_cc - 3 &= \frac{1}{2}(h_c-1)c |*2 \\
\\
h_cc + 7 &= (h_c+3)c \\
h_cc - 3 &= (h_c-1)c \\
\\
h_cc + 7 &= h_cc+3c |-h_cc \\
h_cc - 3 &= h_cc-c |-h_cc \\
\\
7 &= 3c \\
3 &=-c \\
\\
\frac{7}{3} &= c \\
-3 &=c \\
\end{align}
```

$$A = \frac{1}{2}h_cc \tag{113}$$
$$A + 7 = \frac{1}{2}(h_c + 3)c \tag{114}$$
$$A - 3 = \frac{1}{2}(h_c - 1)c \tag{115}$$
$$\tag{116}$$
$$\frac{1}{2}h_cc + 7 = \frac{1}{2}(h_c + 3)c| * 2 \tag{117}$$
$$\frac{1}{2}h_cc - 3 = \frac{1}{2}(h_c - 1)c| * 2 \tag{118}$$
$$\tag{119}$$
$$h_cc + 7 = (h_c + 3)c \tag{120}$$
$$h_cc - 3 = (h_c - 1)c \tag{121}$$
$$\tag{122}$$
$$h_cc + 7 = h_cc + 3c| - h_cc \tag{123}$$
$$h_cc - 3 = h_cc - c| - h_cc \tag{124}$$
$$\tag{125}$$
$$7 = 3c \tag{126}$$
$$3 = -c \tag{127}$$
$$\tag{128}$$
$$\frac{7}{3} = c \tag{129}$$
$$-3 = c \tag{130}$$

In [16]:

```
%%latex
\begin{align}
2a + 2b + 6c - 31 &= 19 | +32\\
a + b + 3c &= 25 \\
-4a + b &= 10 - 3a + c |+3a - c\\
\\
2a + 2b + 6c &= 50 |/2\\
a + b + 3c &= 25 \\
-a + b - c &= 10\\
\\
a + b + 3c &= 25 \\
a + b + 3c &= 25 \\
-a + b - c &= 10\\
\\
```

```
a + b + 3c &= 25 \\
0 &= 0 \\
2b - 2c &= 35\\
\\
b - c &= \frac{35}{2}\\
\\
b &= \frac{35}{2} + c\\
\\
a + \frac{35}{2} + c + 3c &= 25 | -\frac{35}{2} - 4c\\
\\
a &= \frac{15}{2} - 4c\\
\end{align}
```

$$
\begin{aligned}
2a + 2b + 6c - 31 &= 19| + 32 && (131)\\
a + b + 3c &= 25 && (132)\\
-4a + b &= 10 - 3a + c| + 3a - c && (133)\\
& && (134)\\
2a + 2b + 6c &= 50|/2 && (135)\\
a + b + 3c &= 25 && (136)\\
-a + b - c &= 10 && (137)\\
& && (138)\\
a + b + 3c &= 25 && (139)\\
a + b + 3c &= 25 && (140)\\
-a + b - c &= 10 && (141)\\
& && (142)\\
a + b + 3c &= 25 && (143)\\
0 &= 0 && (144)\\
2b - 2c &= 35 && (145)\\
& && (146)\\
b - c &= \frac{35}{2} && (147)\\
& && (148)\\
b &= \frac{35}{2} + c && (149)\\
& && (150)\\
a + \frac{35}{2} + c + 3c &= 25| - \frac{35}{2} - 4c && (151)\\
& && (152)\\
a &= \frac{15}{2} - 4c && (153)
\end{aligned}
$$

In [17]:

```
%%latex
\begin{align}
A*B &= \frac{1}{8} \\
A*\frac{B}{2} &= \frac{1}{16} \\
\frac{A}{B} &= \frac{B}{C} \\
C &= \frac{A}{2} \\
\end{align}
```

$$A * B = \frac{1}{8} \tag{154}$$

$$A * \frac{B}{2} = \frac{1}{16} \tag{155}$$

$$\frac{A}{B} = \frac{B}{C} \tag{156}$$

$$C = \frac{A}{2} \tag{157}$$